[0x0c]

$\oplus$ distribute $\Leftarrow$

$(a \oplus b) \circlearrowleft^{d\text{-bits}}$ $\begin{cases} c = a \\ c = (c \oplus b) \\ c = \circlearrowleft d \text{ bits} \end{cases}$

$\|$

$(a \circlearrowleft^{d\text{-bits}}) \oplus (b \circlearrowleft^{d\text{-bits}})$ $\begin{cases} a \circlearrowleft d \text{ bits} \\ b \circlearrowleft d \text{ bits} \\ a = (a \oplus b) \end{cases}$

xor distributive

```
mov     rcx, rax
xor     rcx, rbx
ror     rcx, 0xd

ror     rax, 0xd
ror     rbx, 0xd
xor     rax, rbx

cmp     rax, rcx
```

rcx = rax

preserve equality

rcx = rax

ie ZF = 1

mov rcx, rax — ie rax → rcx

xor rcx, rbx — store diff bits of rbx & rcx (orig. rax) into rcx

ror rcx, 0xd — "rotate right" rcx (bit preserving, ie LSB → MSB) (the diffs)

ror rax, 0xd — rycle original rax

ror rbx, 0xd — rycle original rbx

xor rax, rbx — same diffs between rax & rbx → rax

cmp rax, rcx → ZF := 1 (as should be ==)

ie rax & rcx end with the same value...

ie xor distributive